

A Brief Introduction to Rietveld Analysis of XRD Patterns

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References:

1. R.A. Young, The Rietveld Method
2. Georg Will, Powder Diffraction: The Rietveld Method and the Two Stage Method to Determine and Refine Crystal Structures from Powder Diffraction Data

Timeline

- 1966 : 7th Congress of the IUCr in Moscow
- 1969: publication in J. Appl. Crystallogr.
- 1977: Acceptance for X-ray diffraction
- 1983: Quantitative Phase Analysis



Dr. Hugo M. Rietveld

Difficulties with Powder Diffraction

- Systematic overlapping of diffraction peaks due to symmetry conditions, for example in cubic space groups
- Accidental overlapping because of limited experimental resolution
- Considerable background difficult to define with accuracy
- Non-random distribution of the crystallites in the specimen, generally known as preferred orientation.

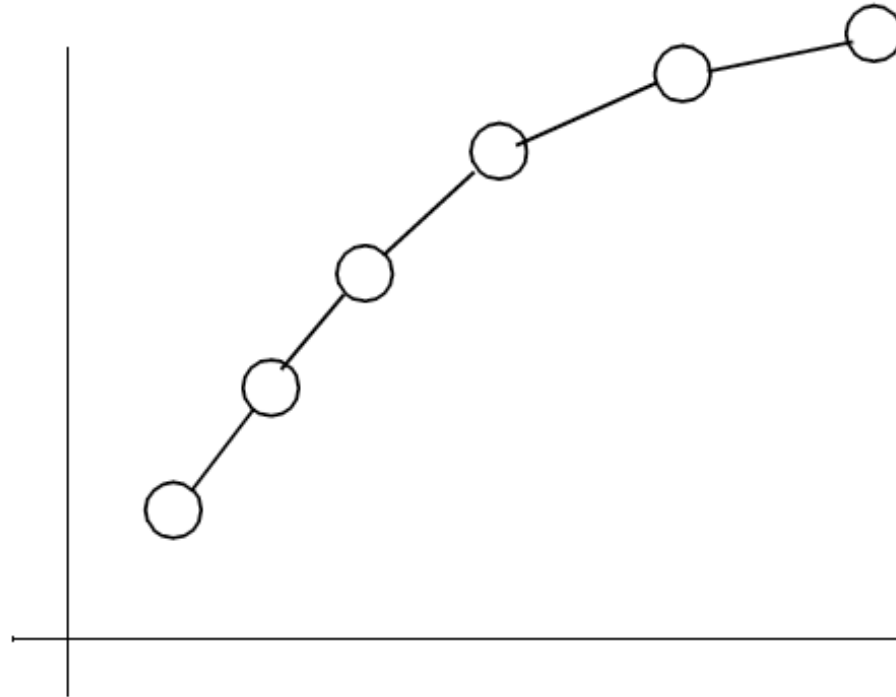
What is Rietveld analysis?

- The Rietveld method refines user-selected parameters to minimize the difference between an experimental pattern (observed data) and a model based on the hypothesized crystal structure and instrumental parameters (calculated pattern)
- Full profile fitting
- Using crystallographic constraints
 - Lattice parameters and space group to constrain peak positions
 - Crystal structure to constrain peak intensities

Least squares method

Curve Fitting

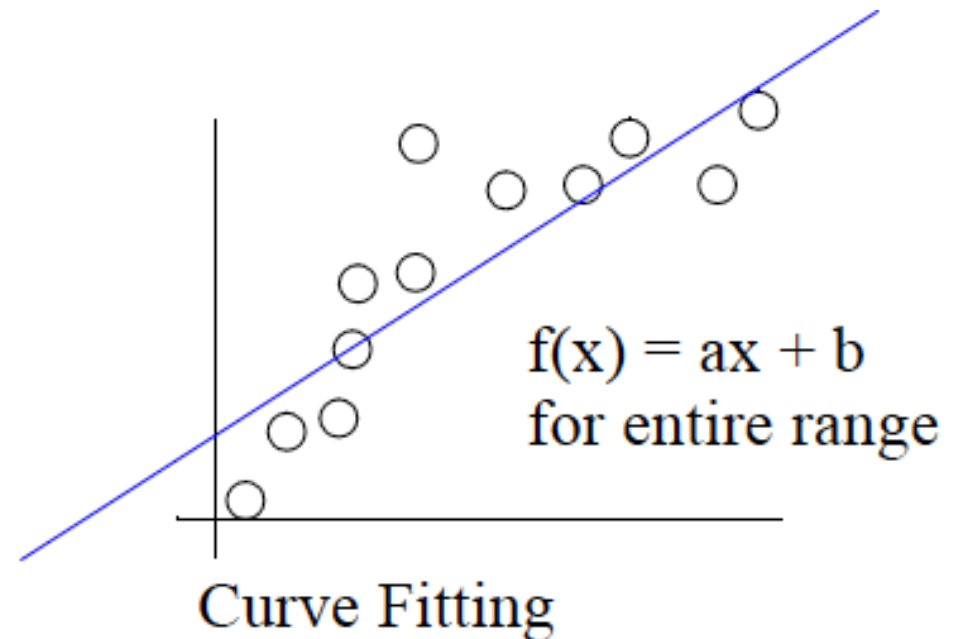
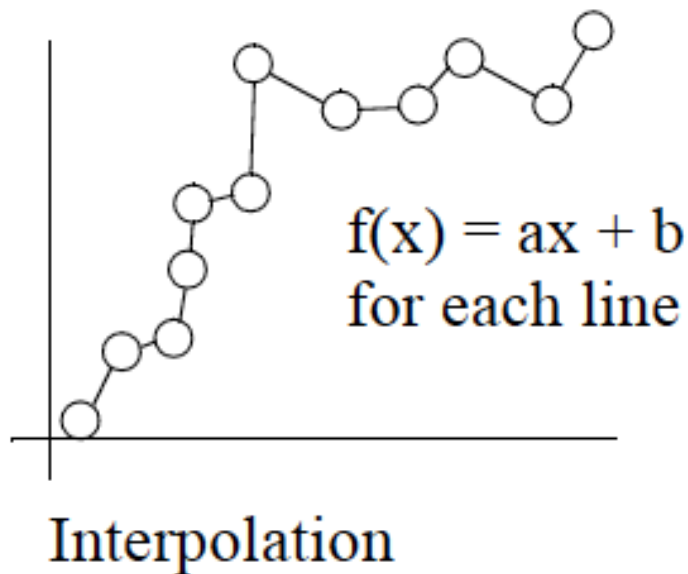
Interpolation



- Interpolation
 - connect the data-dots
- If data is reliable, we can plot it and connect the dots
- This is piece-wise, linear interpolation

Curve fitting

- Depicting the trend in the data variation by assigning a single function to represent the data across its entire range



A straight line function
 $f(x) = ax + b$

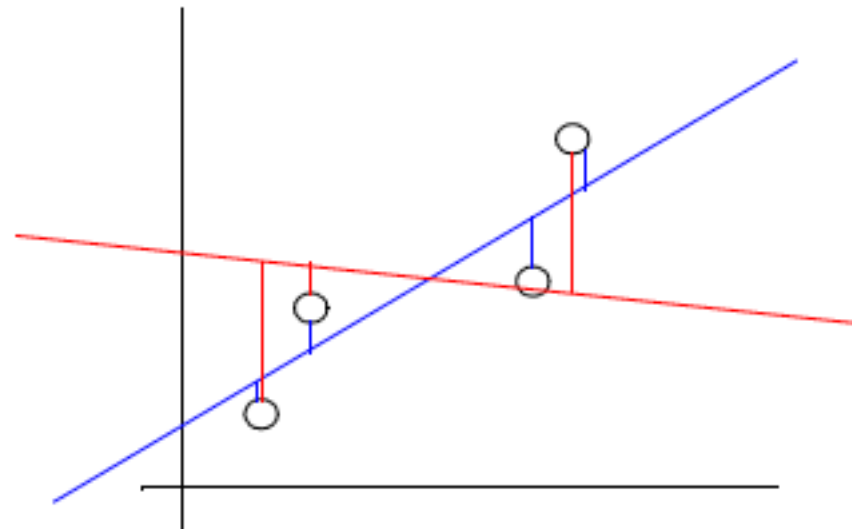
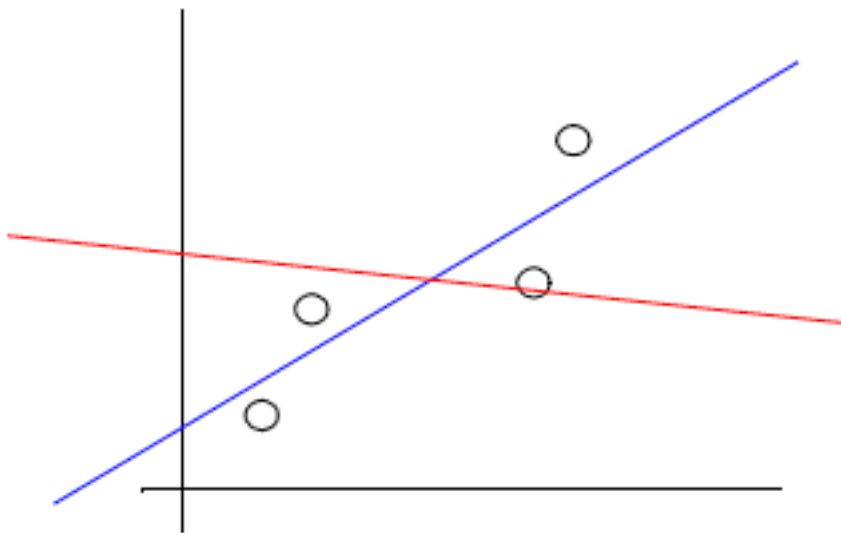
The goal is to identify the coefficients 'a' and 'b' such that $f(x)$ 'fits' the data well

Linear Curve Fitting or Regression

A straight line function $f(x) = ax + b$

How can we pick the coefficients that best fits the line to the data?

First question: What makes a particular straight line a ‘good’ fit?



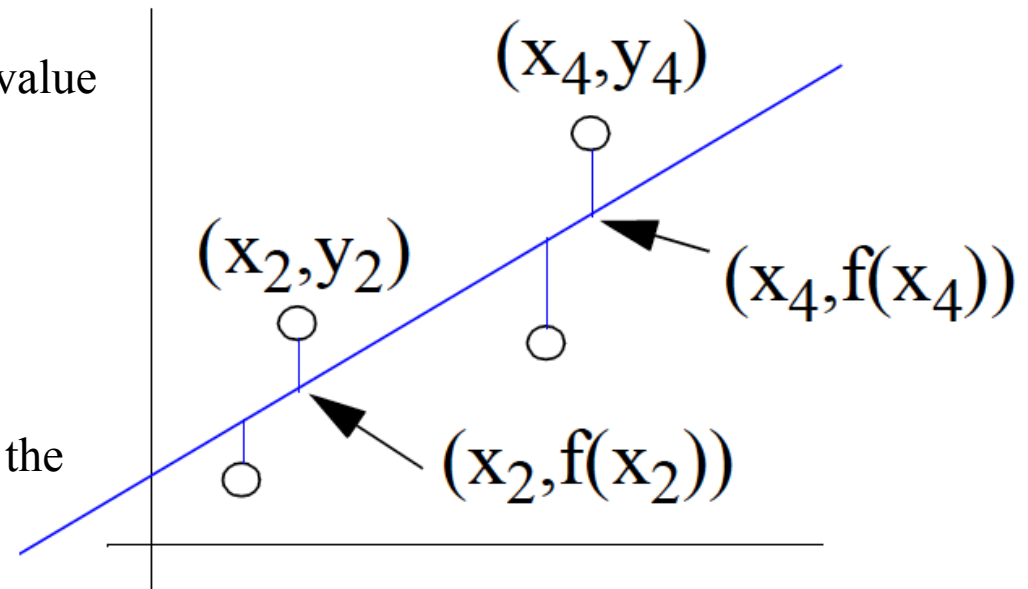
Quantifying error in a curve fit

Assumptions:

- 1) positive or negative error have the same value
(data point is above or below the line)
- 2) Weight greater errors more heavily

Square the distance

Denote data values as (x, y) and points on the fitted line as $(x, f(x))$



sum the error at the four data points

$$\begin{aligned} \text{Error} &= \sum \left(d_i^2 \right) = \left(y_1 - f(x_1) \right)^2 + \left(y_2 - f(x_2) \right)^2 + \left(y_3 - f(x_3) \right)^2 + \left(y_4 - f(x_4) \right)^2 \\ &= \sum_l^n \left(y_i - a(x_i) - b_i \right)^2 \end{aligned}$$

The 'best' line has minimum error between line and data points.

This is called the **least squares approach**, since we minimize the square of the error.

$$\frac{\partial err}{\partial a} = -2 \sum_{i=1}^n x_i(y_i - ax_i - b) = 0$$

$$\frac{\partial err}{\partial b} = -2 \sum_{i=1}^n (y_i - ax_i - b) = 0$$

$$a \sum x_i^2 + b \sum x_i = \sum (x_i y_i)$$

$$a \sum x_i + b * n = \sum y_i$$

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \end{bmatrix}$$

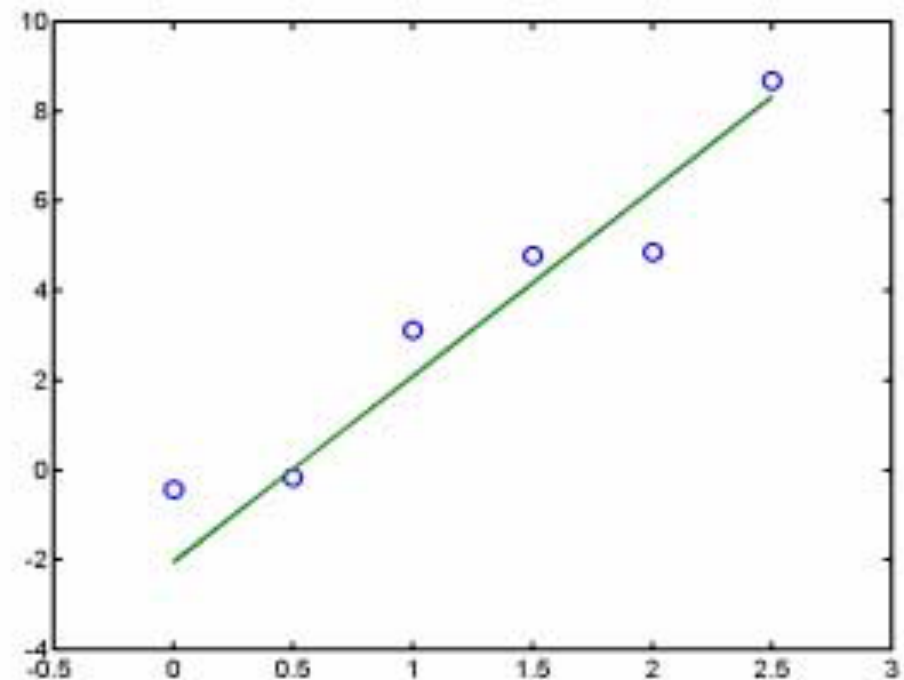
$$A \quad X = B$$

Example:

$$x = [0 \ 0.5 \ 1 \ 1.5 \ 2 \ 2.5], \quad y = [-0.4326 \ -0.1656 \ 3.1253 \ 4.7877 \ 4.8535 \ 8.6909]$$

$$\begin{bmatrix} 6 & 7.5 \\ 7.5 & 13.75 \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} 20.8593 \\ 41.6584 \end{bmatrix}, \quad \begin{bmatrix} b \\ a \end{bmatrix} = \text{inv} \begin{bmatrix} 6 & 7.5 \\ 7.5 & 13.75 \end{bmatrix} * \begin{bmatrix} 20.8593 \\ 41.6584 \end{bmatrix}, \quad \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} -0.975 \\ 3.561 \end{bmatrix}$$

$$f(x) = 3.561x - 0.975$$



So...what do we do when a straight line is not suitable for the data set?

Nonlinear Curve Fitting

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_jx^j = a_0 + \sum_{k=1}^j a_kx^k$$

Just as was the case for linear regression:

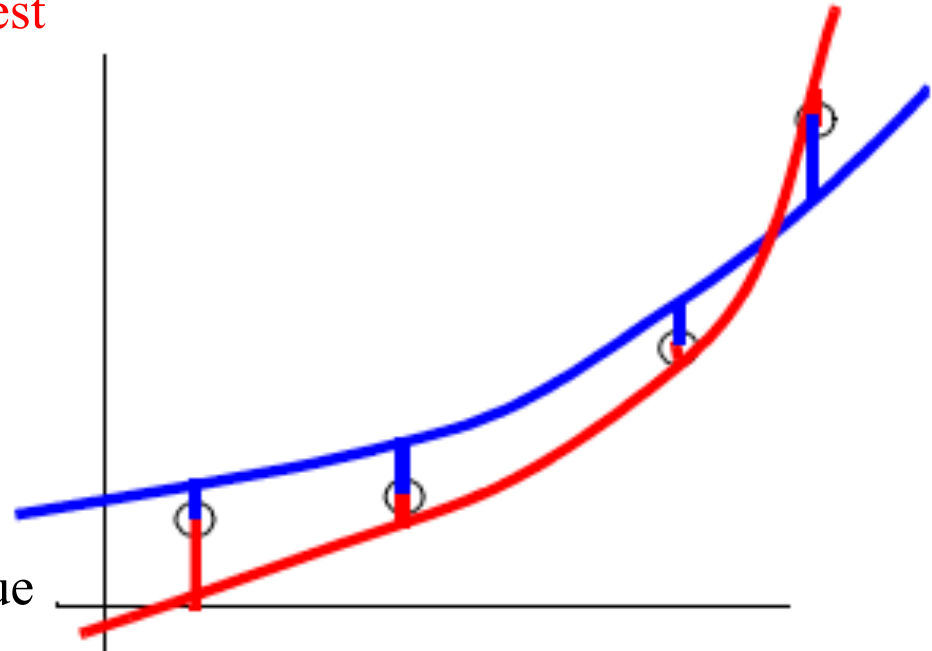
How can we pick the coefficients that best fit the curve to the data?

The curve that gives minimum error between data and the fit is 'best'

Quantify the error for these two second order curves...

- Add up the length of all the red and blue vertical lines

- pick curve with minimum total error



$$Error = \sum (d_i^2) = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2 + (y_4 - f(x_4))^2$$

$$f(x) = a_o + a_1x + a_2x^2 + a_3x^3 + + a_jx^j = a_o + \sum_{k=1}^j a_k x^k$$

$$Error = \sum_{i=1}^n \left(y_i - \left(a_o + a_1x_i + a_2x_i^2 + a_3x_i^3 + + a_jx_i^j \right) \right)^2$$

$$= \sum_{i=1}^n \left(y_i - \left(a_o + \sum_{k=1}^j a_k x_i^k \right) \right)^2$$

n : # of data points given

i - the current data point being summed

j - the polynomial order

$$\frac{\partial err}{\partial a_0} = -2 \sum_{i=1}^n \left(y_i - \left(a_0 + \sum_{k=1}^j a_k x^k \right) \right) = 0$$

$$\frac{\partial err}{\partial a_1} = -2 \sum_{i=1}^n \left(y_i - \left(a_0 + \sum_{k=1}^j a_k x^k \right) \right) x = 0$$

$$\frac{\partial err}{\partial a_2} = -2 \sum_{i=1}^n \left(y_i - \left(a_0 + \sum_{k=1}^j a_k x^k \right) \right) x^2 = 0$$

⋮

⋮

$$\frac{\partial err}{\partial a_j} = -2 \sum_{i=1}^n \left(y_i - \left(a_0 + \sum_{k=1}^j a_k x^k \right) \right) x^j = 0$$

$$\begin{bmatrix}
 n & \sum x_i & \sum x_i^2 & \dots & \sum x_i^j \\
 \sum x_i & \sum x_i^2 & \sum x_i^3 & \dots & \sum x_i^{j+1} \\
 \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \dots & \sum x_i^{j+2} \\
 \vdots & \vdots & \vdots & & \vdots \\
 \sum x_i^j & \sum x_i^{j+1} & \sum x_i^{j+2} & \dots & \sum x_i^{j+j}
 \end{bmatrix}
 \begin{bmatrix}
 a_0 \\
 a_1 \\
 a_2 \\
 \vdots \\
 a_j
 \end{bmatrix}
 =
 \begin{bmatrix}
 \sum y_i \\
 \sum (x_i y_i) \\
 \sum (x_i^2 y_i) \\
 \vdots \\
 \sum (x_i^j y_i)
 \end{bmatrix}$$

Example: Fitting a 2nd order polynomial

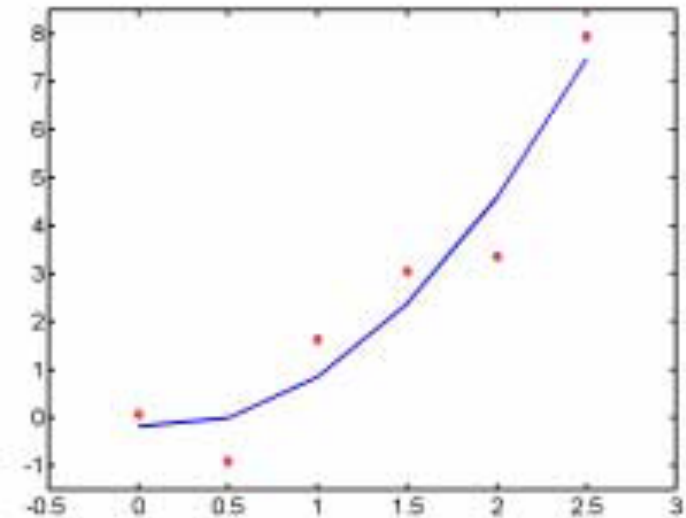
$$x = [0 \ 0.1 \ 1 \ 1.5 \ 2 \ 2.5], y = [0.0674 \ -0.9156 \ 1.6253 \ 3.0377 \ 3.3535 \ 7.9409]$$

For a 2nd order polynomial

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \text{inv} \begin{bmatrix} 6 & 7.5 & 13.75 \\ 7.5 & 13.75 & 28.125 \\ 13.75 & 28.125 & 61.1875 \end{bmatrix} * \begin{bmatrix} 15.1093 \\ 32.2834 \\ 71.276 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -0.1812 \\ -0.3221 \\ 1.3537 \end{bmatrix}$$

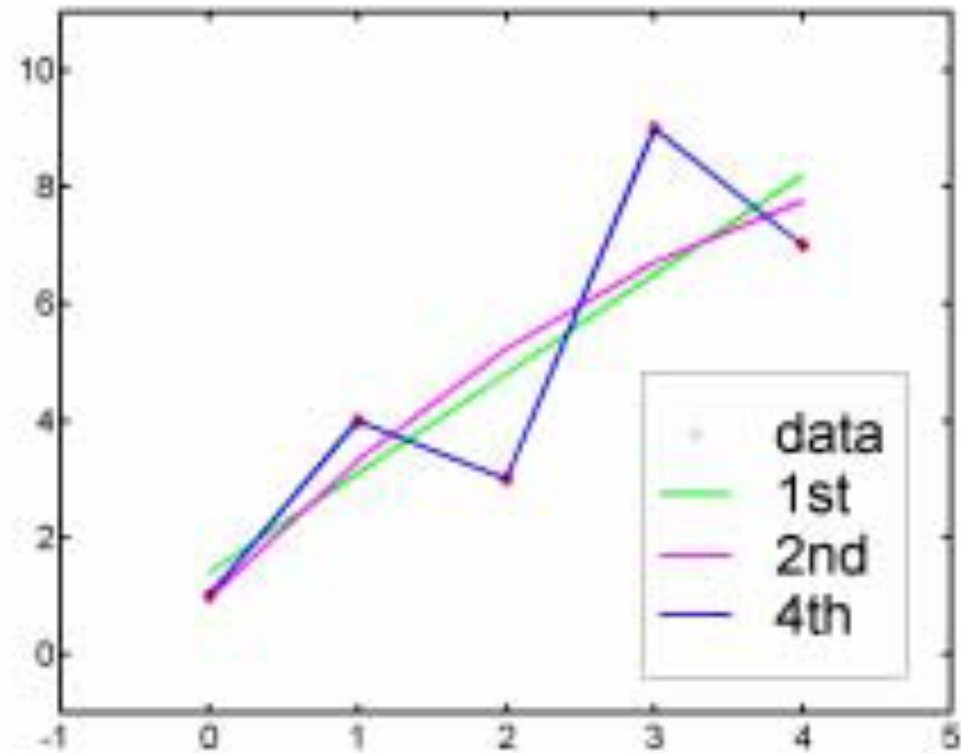


$$f(x) = -0.1812 - 0.3221x + 1.3537x^2$$

Fitting considerations

Pick a polynomial form at least several orders lower than the number of data points.

Start with linear and add order until trends are matched.



What Rietveld can do?

- Analysis of the whole diffraction pattern
 - Profile fitting is included
 - Not only the integrated intensities
- Refinement of the structure parameters from diffraction data
 - Quantitative phase analysis (crystalline and amorphous)
 - Lattice parameters
 - Atomic positions and occupancies
 - Temperature vibrations (isotropic and anisotropic)
- Other information
 - Grain size and micro micro-strain (isotropic and anisotropic)
 - Stacking and twin faults
 - Magnetic moments (neutrons)
- Not intended for the structure solution
 - The structure model must be known before starting the Rietveld refinement

Requirements of Rietveld Method

- High quality experimental diffraction pattern
- A structure model that makes physical and chemical sense
- Suitable peak and background functions

Principle of Rietveld Refinement

Calculated intensity at point i of the diagram

$$y_{ic} = y_{ib} + \sum_{\Phi} S_{\Phi} \sum_k G_{\Phi} (2\theta_i - 2\theta_k) I_k$$

G : normalized profile shape function

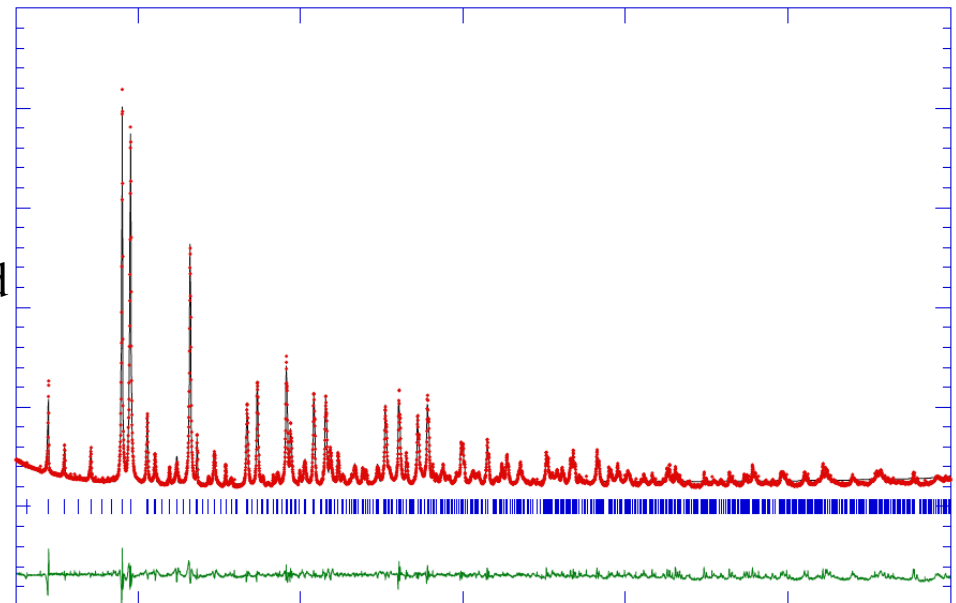
I : intensity of the k^{th} reflection

S : scale factor of phase Φ Summation performed over all phases Φ , and over all reflections $\mathbf{k}$ contributing to the respective point.

y_{ib} : background

Intensity of the Bragg reflections

$$I_k = m_k L_k |F_k|^2 P_k A_k$$



m_k : multiplicity of k

L_k : Lorentz-polarization factor

$|F_k|^2$: structure factor

P_k : preferred orientation factor

A_k : absorption factor

Goal

- To minimize the residual function

$$\sum_i w_i \left(y_i^{obs} - y_i^{calc} \right)^2$$

where $w_i = 1/y_i^{obs}$

y_i^{obs} : observed intensity at the i^{th} step

y_i^{calc} : calculated intensity at the i^{th} step

$$I_i^{calc} = S_F \sum_{j=1}^{Nphases} \frac{f_j}{V_j^2} \sum_{k=1}^{Mpeaks} L_k |F_{k,j}|^2 S_j (2\theta_i - 2\theta_{k,j}) P_{k,j} A_j + bkg_i$$

Scale Factor

S_F : beam intensity

f_j : volume fraction

V_j : cell volume

Structure Factor

Multiplicity of k-th
reflection (m_k)

Temperature factor

Volume
absorption

Preferred
orientation

Lorentz--Polarization factor

- Geometry
- monochromator (angle θ)
- Detector
- beam size/sample volume
- sample positioning (angular)

i : steps

j : number of phases

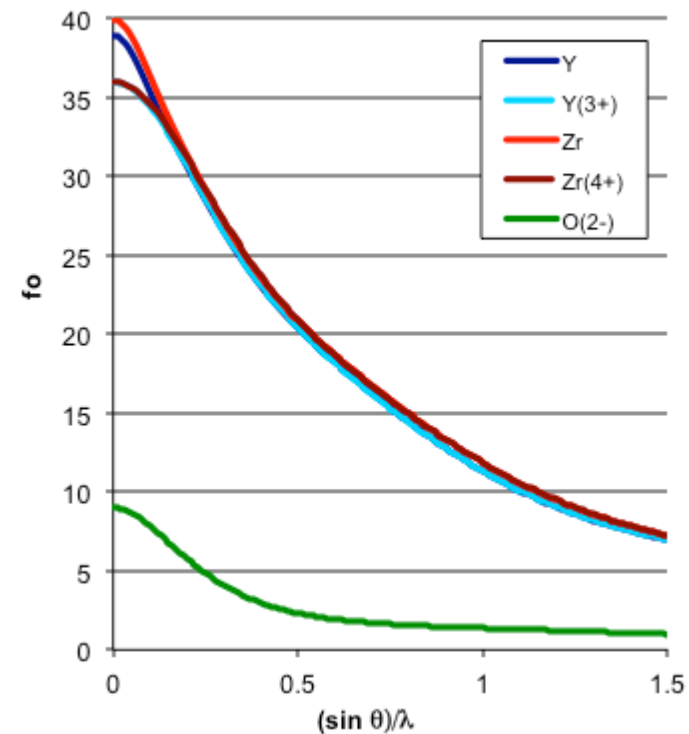
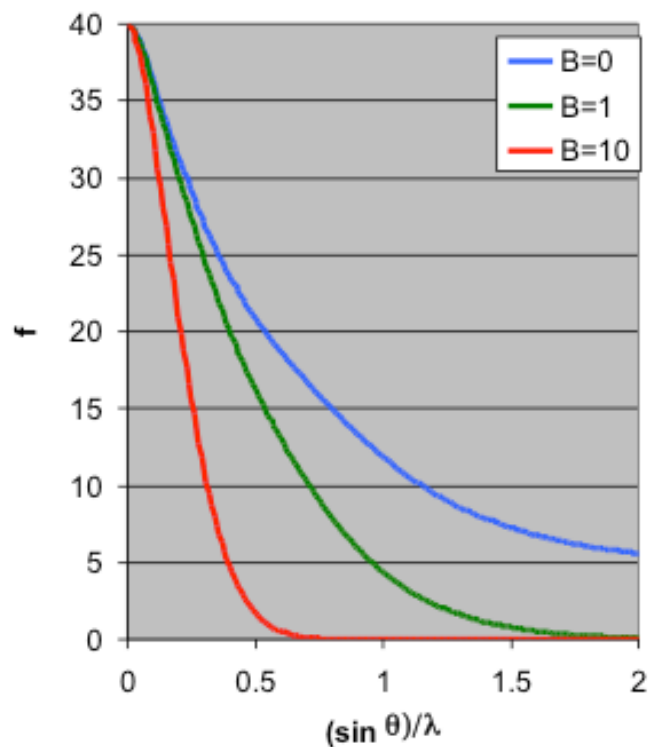
k : k^{th} reflection

Structure Factor

$$F_{hkl} = \sum_{j=1}^m N_j f_j \exp[2\pi i(hx_j + ky_j + lz_j)]$$

$$|f|^2 = \left(f_0 \exp\left[\frac{-B \sin^2 \theta}{\lambda^2}\right] + \Delta f' \right)^2 + (\Delta f'')^2$$

B: Debye Waller factor



With a crystal structure to solve Bragg's Law and F_{hkl} , you can calculate the diffraction pattern from an ideal crystal

- Ideal powder diffraction patterns can be simulated if you know ...
 - space group symmetry
 - unit cell dimensions
 - atom types
 - relative coordinates of atoms in unit cell
 - atomic site occupancies
 - atomic thermal displacement parameters
- ... for each and every phase in the sample

Where to Get Crystal Structures

- publications
- Commercial Databases
 - ICDD PDF4+ Inorganic
 - Inorganic Crystal Structure Database (ICSD)
 - Linus Pauling File (LPF)
 - this is included in ICDD PDF4+ Inorganic
 - NIST Structural Database (metals, alloys, intermetallics)
 - CCDC Cambridge Structure Database (CSD) (organic materials)
 - Available online <http://beta-www.ccdc.cam.ac.uk/pages/Home.aspx>
 - MIT Site license available for download from IS&T website
- Free Online Databases
 - Crystallography Open Database <http://www.crystallography.net/>
 - ICSD- 4% available as demo at <http://icsd.ill.eu/icsd/index.html>
 - Mincryst <http://database.iem.ac.ru/mincryst/index.php>
 - American Mineralogist http://www.minsocam.org/MSA/Crystal_Database.html
 - WebMineral <http://www.webmineral.com/>
 - Protein Data Bank <http://www.rcsb.org/pdb/home/home.do>
 - Nucleic Acid Database <http://ndbserver.rutgers.edu/>
 - Database of Zeolite Structures <http://www.iza-structure.org/databases/>

Lorentz polarization factor

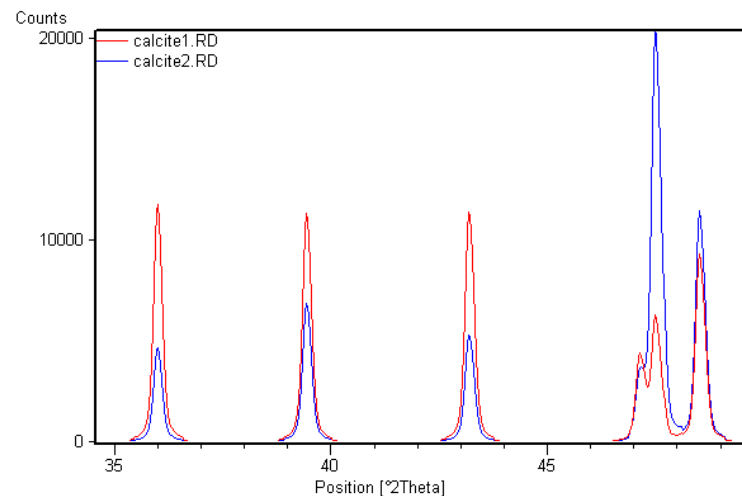
- The Lorentz factor
 - Different planes remain in the diffracting condition over a different angular range
 - high angle planes remain in the diffracting position longer, increasing their apparent intensity
 - $1/\sin^2\theta\cos\theta$ relationship for Bragg-Brentano
 - varies with geometry of the diffractometer
 - Polarization
 - Scattering of light by the atom polarizes the X-rays
 - this causes the scattered intensity to vary as $1+\cos^2(2\theta)/2$
 - Monochromator Polarization
 - The monochromator uses diffraction from a crystal to remove unwanted wavelengths of radiation.
 - This diffraction will also polarize the X-ray beam, causing intensity to vary as $\cos^2(2\theta)$
 - POL is the correction factor for the monochromator
- the correct POL value can be found in HSP Help
- go to the Help Index, type in POL, and the first option describes the Lorentz polarization factor

$$L_p = \frac{1 + POL \cos^2 2\theta}{\sin^2 \theta \cos \theta}$$

Monochromator	POL
K-beta Filter	1
Cu diffracted beam graphite (point detector)	0.8
Cu X'Celerator	1.25
Cr diffracted beam (point detector)	0.588

Modeling Preferred Orientation

- The default calculation assumes that there are an equal number of crystallites contributing to every diffraction peak
 - the orientation of the crystallites is random and there are a statistically relevant number of grains
- Preferred crystallographic orientation produces a non-random distribution in the orientation of the crystallites
 - the observed diffraction peak intensities will vary systematically



The March distribution function

- Empirically models the preferred orientation effect
- A model for fitting preferred orientation of cylinder or needle shaped crystals.

$$P_k = \left(r^2 \cos^2 \alpha_k + r^{-1} \sin^2 \alpha_k \right)^{-3/2}$$

α_k is the angle between the preferred orientation vector and the normal to the planes generating the diffracted peak

r is a refinable parameter in the Rietveld method

Additional Intensity Factors

- Absorption Correction

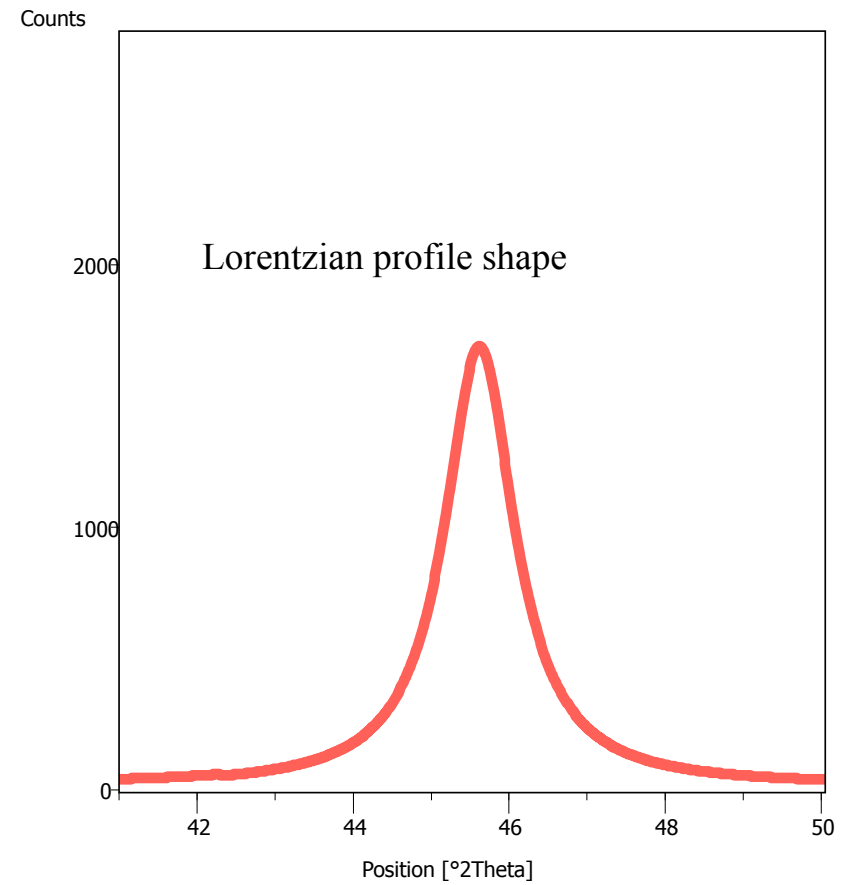
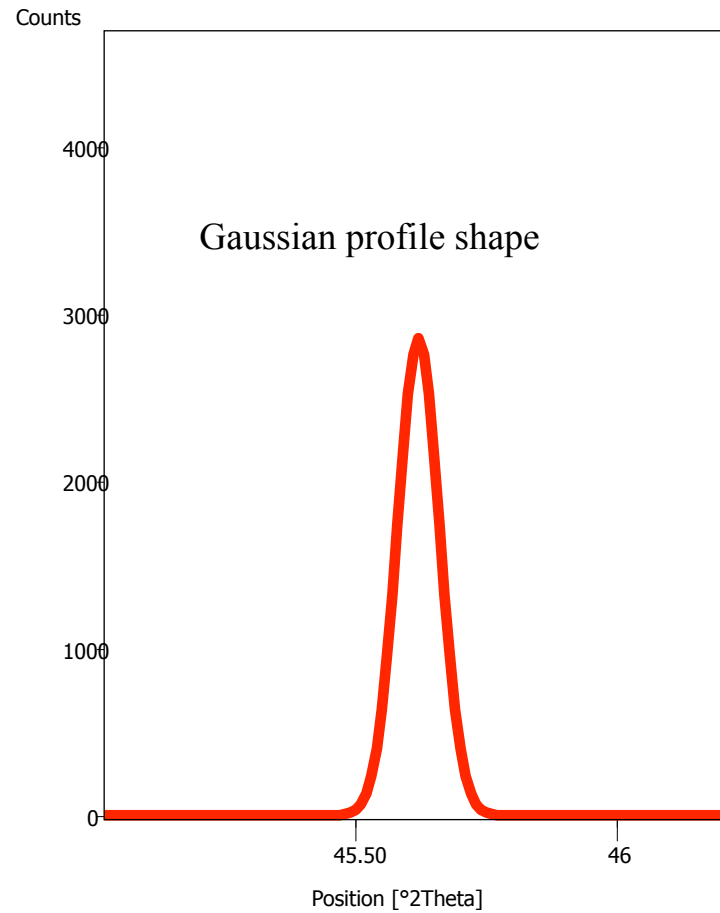
- accounts for the transmission and absorption of the X-rays through the irradiated volume of the sample.
- depends on sample geometry: flat plate or cylindrical
- depends on the mass absorption coefficient of the sample
 - significant for cylindrical or transmission samples with a high sample absorption
 - significant for a flat plate sample with a low sample absorption

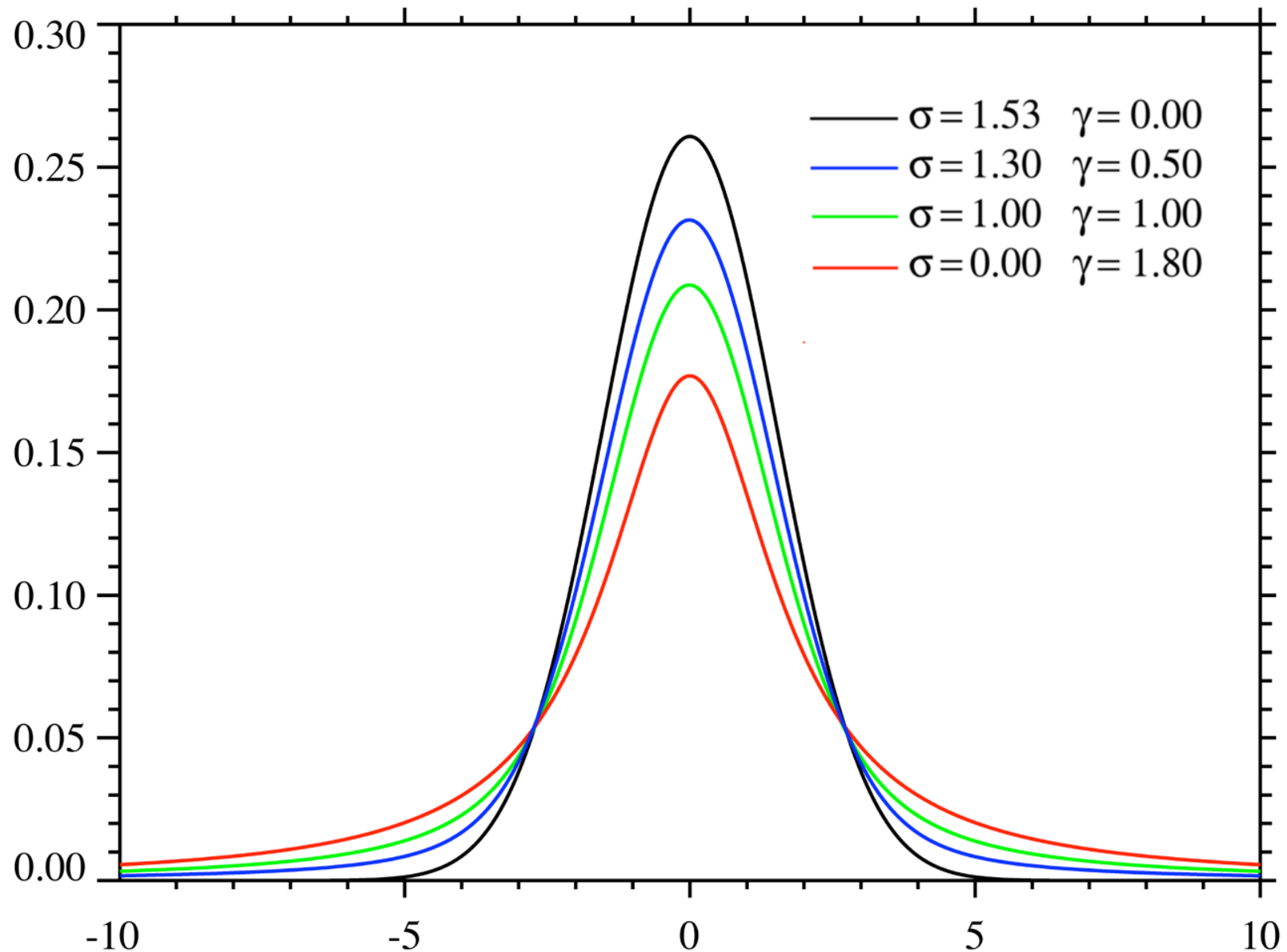
- Extinction Correction

- Secondary diffraction of scattered X-rays in a large perfect crystal can decrease the observed intensity of the most intense peaks
- scattering by uppermost grains limits the penetration depth of a majority of X-rays, causing a smaller irradiated volume
- want “ideally imperfect” crystals– grains with enough mosaicity to limit extinction
- better to reduce extinction by grinding the powder to a finer size

- These two corrections are rarely used. Caution is advised. Always check to make sure that the correction values are physically meaningful.

Profile shape function





Plot of the centered Voigt profile for four cases. Each case has a full width at half-maximum of very nearly 3.6. The black and red profiles are the limiting cases of the Gaussian ($\gamma = 0$) and the Lorentzian ($\sigma = 0$) profiles respectively.

Profile shape functions

Gaussian

$$G = I_0 \exp \left(-\ln 2 \left(\frac{2\Theta - 2\Theta_0}{\omega} \right)^2 \right)$$

Lorentzian

$$L = I_0 \left(1 + \left(\frac{2\Theta - 2\Theta_0}{\omega} \right)^2 \right)^{-n} \quad n = 1; 1.5; 2$$

Pseudo-Voigt

$$V = \eta L + (1 - \eta)G \quad (0 \leq \eta \leq 1)$$

Pearson VII

$$P = I_0 \left(1 + \left(\frac{2\Theta - 2\Theta_0}{ma^2} \right)^2 \right)^{-m}$$

Parameter: $2\Theta_0$ = peak position; I_0 = peak intensity; ω = $FWHM/2$;
 m = shape parameter.

Peak width

Peak half width

$$H_k^2 = U \tan^2 \theta + V \tan \theta + W$$

U, V and W are refinable parameters

Background

Background function is a fifth order polynomial.

$$bkg(2\theta_i) = \sum_{n=0}^{N_b} a_n (2\theta_i)^n$$

N_b : polynomial degree

a_n : polynomial coefficients

Quality of Refinement

- Weighted sum of squares

$$WSS = \sum_{i=1}^N \left[w_i \left(I_i^{\text{exp}} - I_i^{\text{calc}} \right) \right]^2 \quad \text{where } w_i = \frac{1}{\sqrt{I_i^{\text{exp}}}}$$

- R-structure factor

$$R_F = \frac{\sum \left| \left(I_K^{\text{obs}} \right)^{1/2} - \left(I_K^{\text{calc}} \right)^{1/2} \right|}{\sum \left(I_K^{\text{obs}} \right)^{1/2}}$$

- R-Bragg factor

$$R_B = \frac{\sum \left| I_K^{\text{obs}} - I_K^{\text{calc}} \right|}{\sum I_K^{\text{obs}}}$$

I_K : intensity of K^{th} reflection at the end of refinement cycles

Quality of Refinement

- R - weighted pattern (R_{wp})

$$R_{wp} = \left[\frac{\sum w_i (y_i^{obs} - y_i^{calc})^2}{\sum w_i (y_i^{obs})^2} \right]^{1/2}$$

y_i : gross intensity at i^{th} step

- Goodness of Fit (GoF) or χ^2

$$\chi^2 = \frac{R_{wp}}{R_{exp}}$$

$$R_{exp} = \left[(N - P) / \sum w_i (y_i^{obs})^2 \right]^{1/2}$$

N =number of points
 P =number of parameters

Why the Rietveld refinement is widely used?

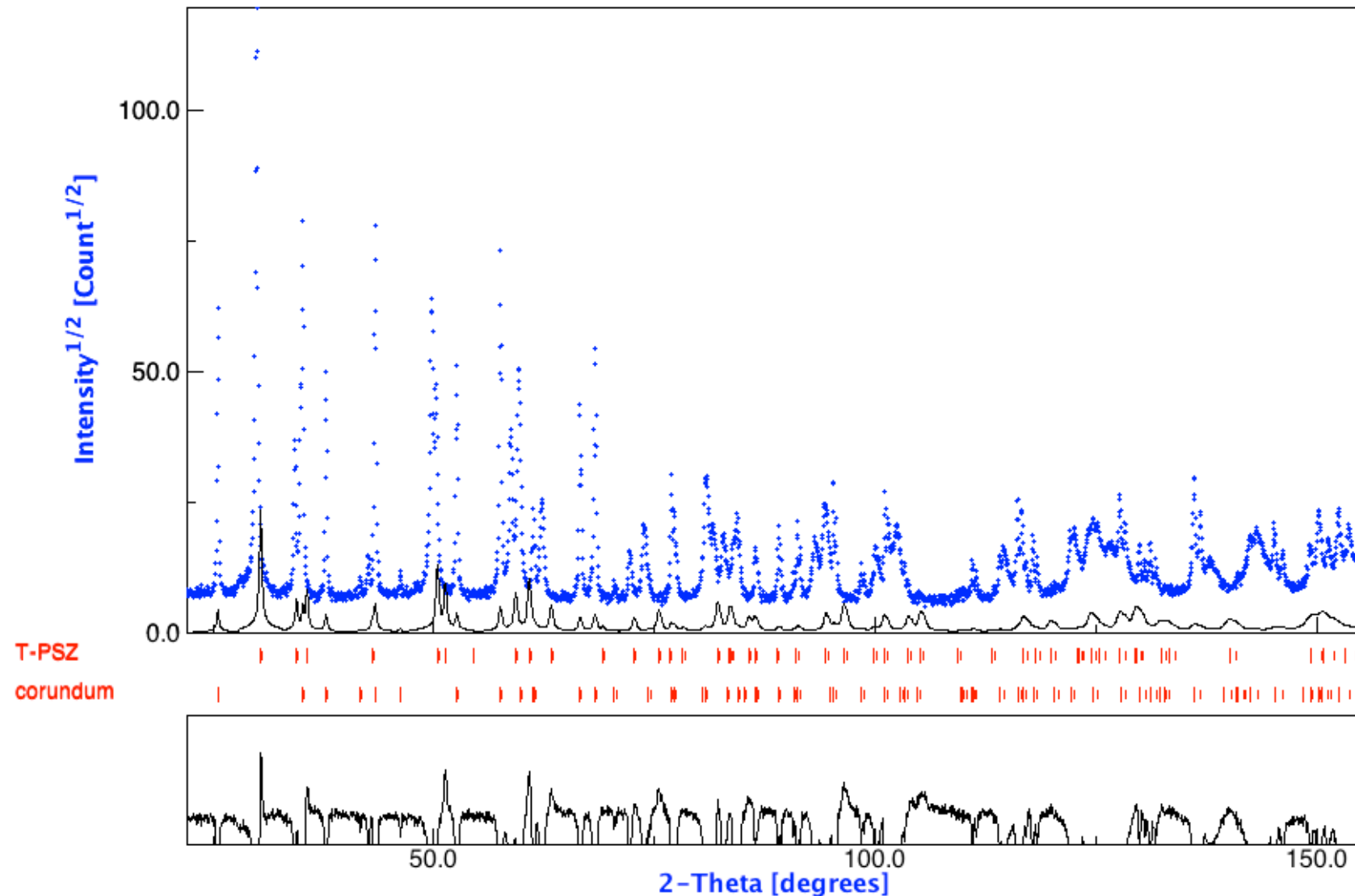
- Pros
 - It uses directly the measured intensities points
 - It uses the entire spectrum (as wide as possible)
 - Less sensible to model errors
 - Less sensible to experimental errors
- Cons
 - It requires a model
 - It needs a wide spectrum
 - Rietveld programs are not easy to use
 - Rietveld refinements require some experience (1-2 years?)
- Can be enhanced by:
 - More automatic/expert mode of operation
 - Better easy to use programs

Rietveld Procedure

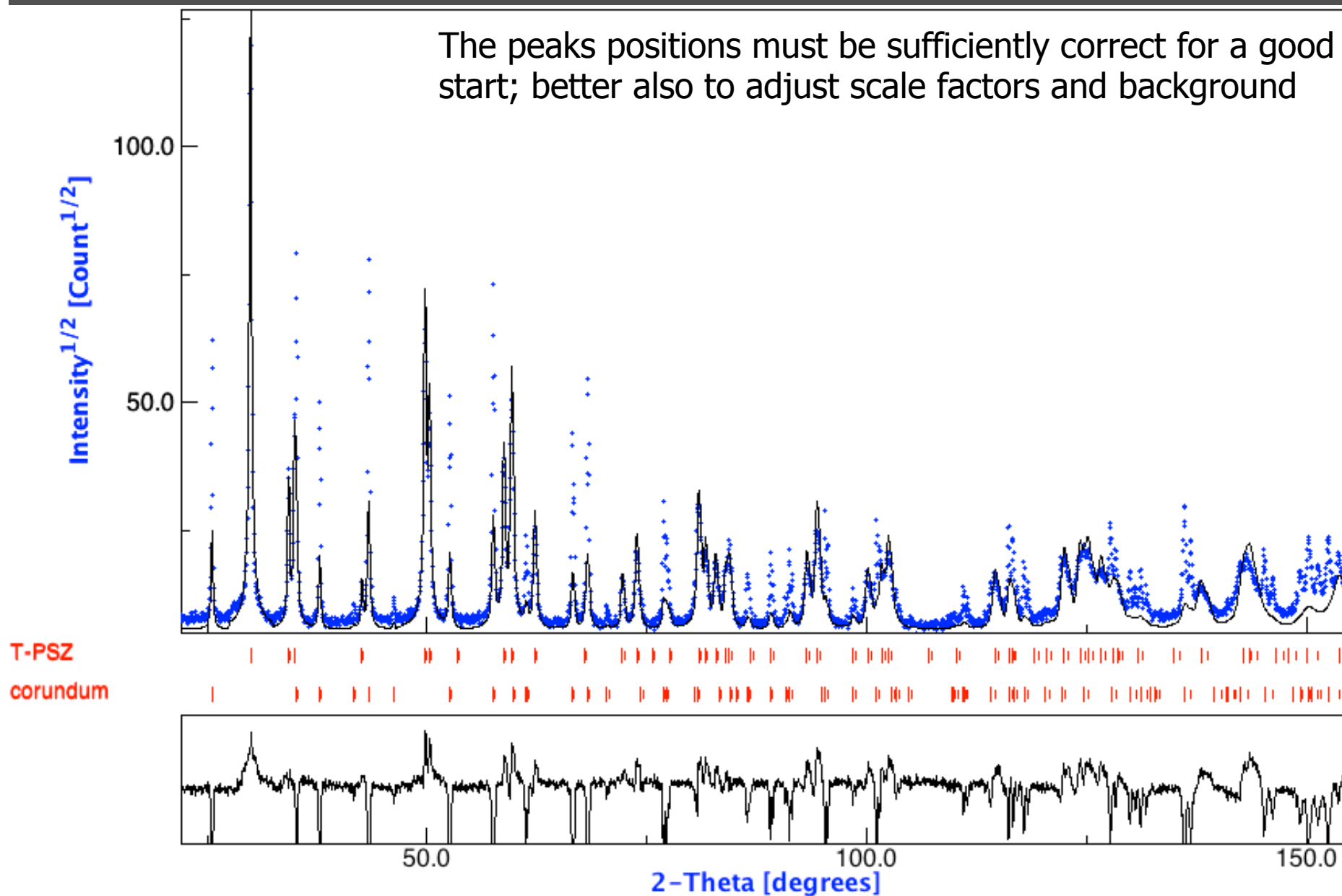
- Experiment:
 - choose the correct instrument/s
 - select the experiment conditions
 - prepare the sample and collect the pattern/s
- Analysis:
 - verify the data quality and perform the qualitative analysis
- Rietveld refinement:
 - Choose appropriate program (GSAS, FullProf or Maud)
 - load or input the phases in the sample
 - adjust manually some parameters (cell, intensities, background)
 - refine overall intensities and background
 - refine peaks positions
 - refine peaks shapes
 - refine structures
 - Assess the results

Example: 1. Defining the phases

We need to specify which phases we will work with (databases)

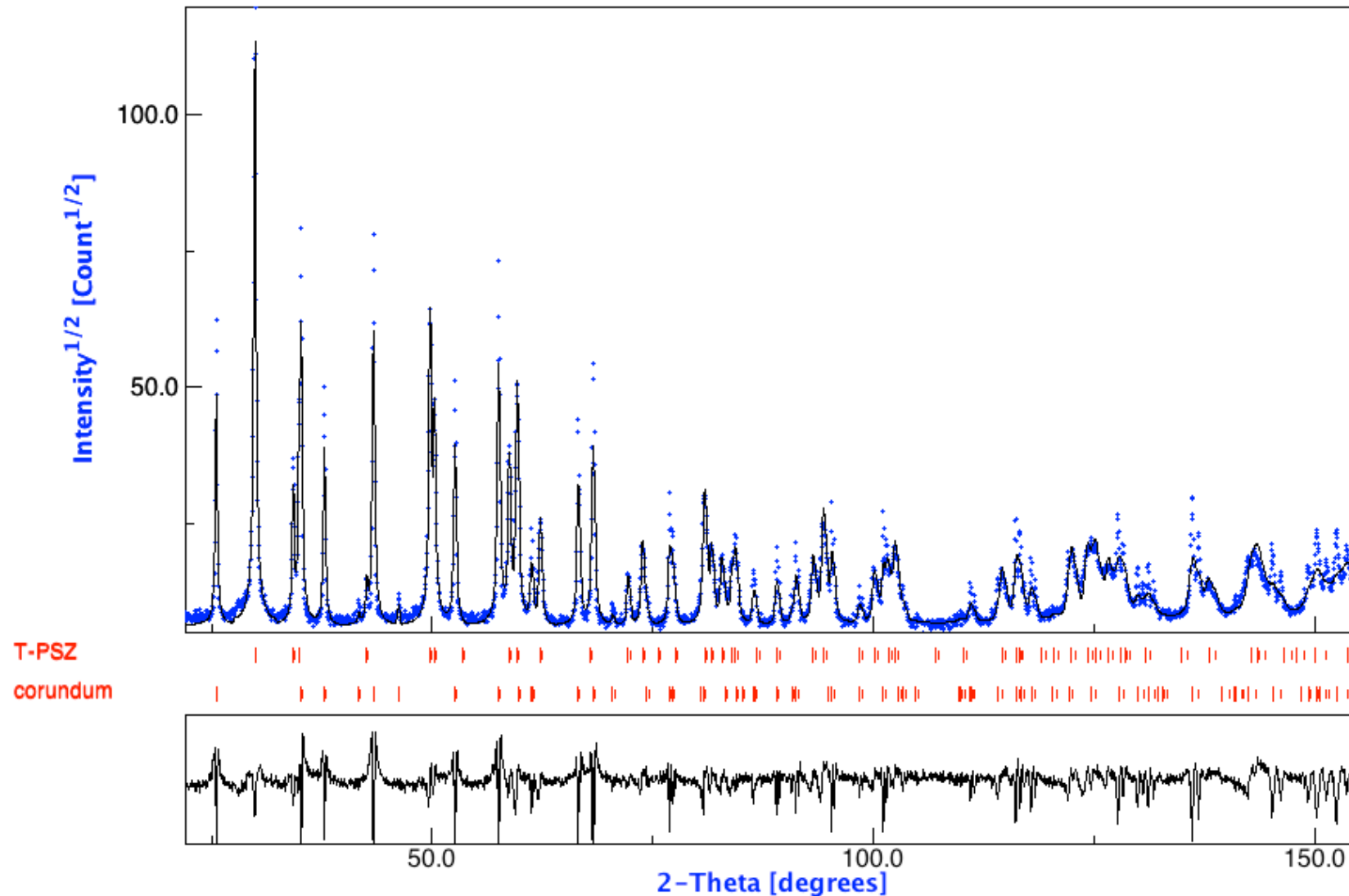


2. Adjusting manually: cell parameters, intensities



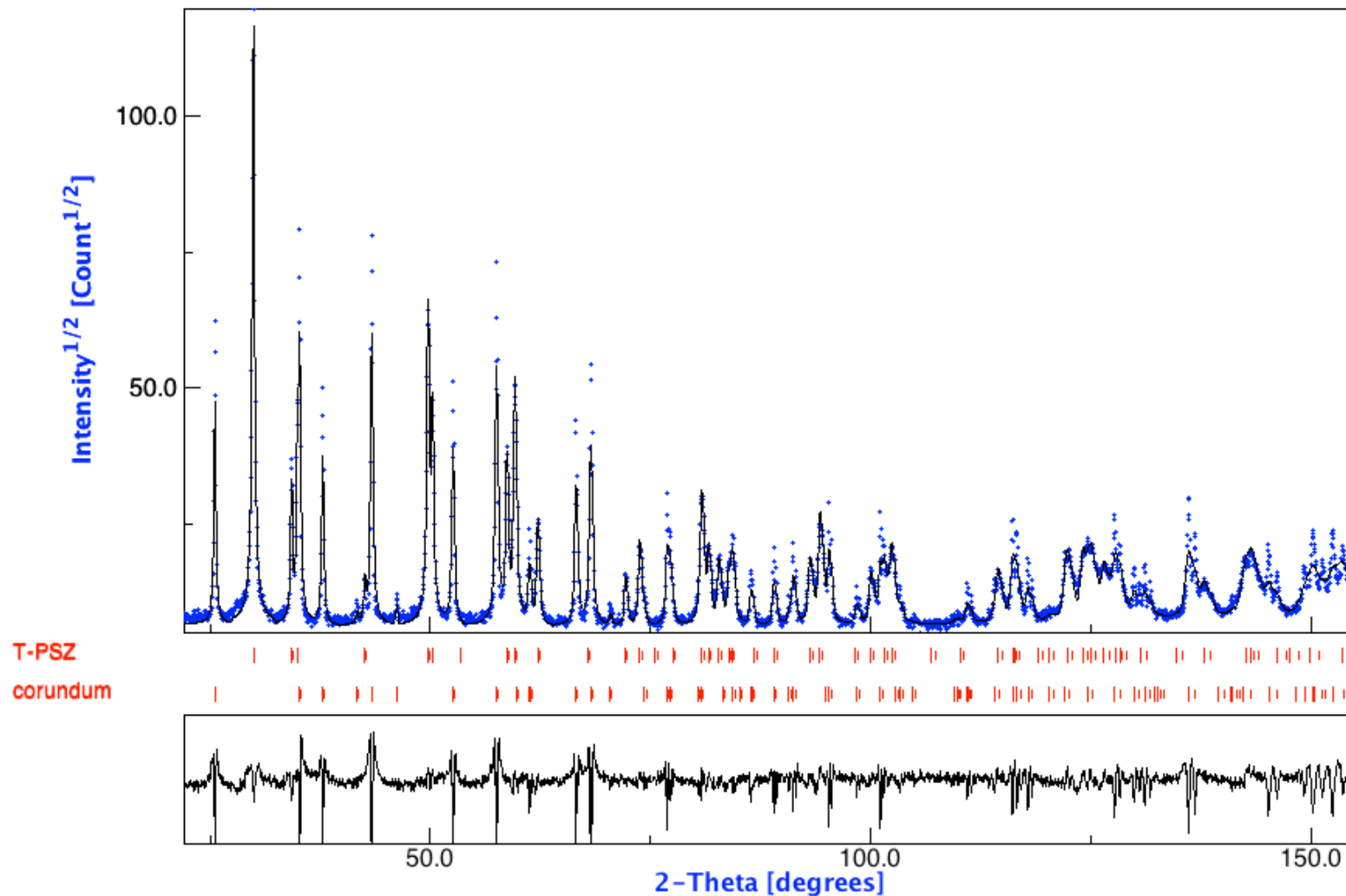
3. Refining scale factors and background

After 5 iterations the R_{wp} is 26.5 %; intensities look better; we use only one overall B factor for all atoms.



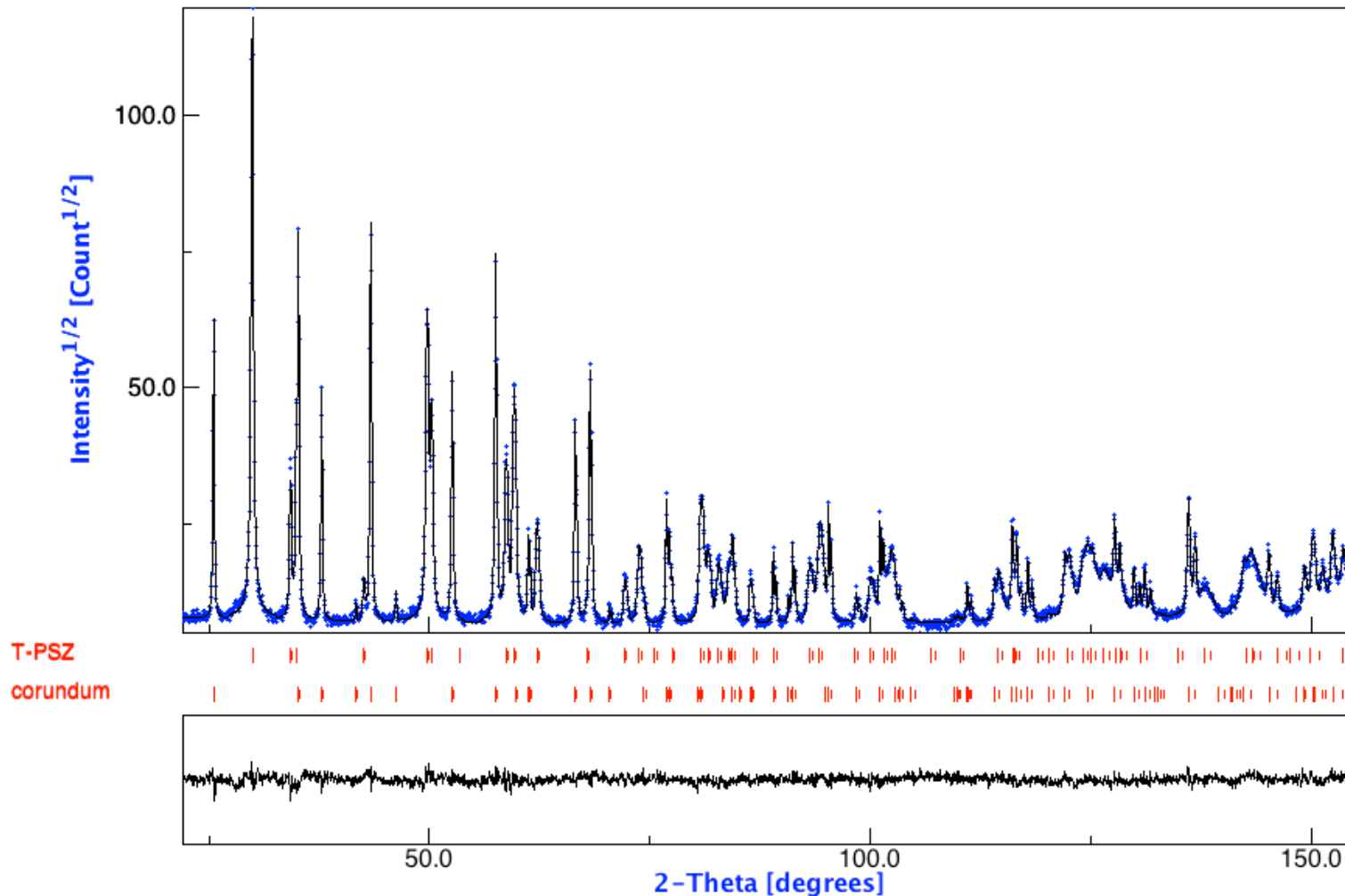
4. Peaks positions

Adding to refinement cell parameters and 2θ displacement; R_{wp} now is at 24.8%; major problems are now peaks shapes



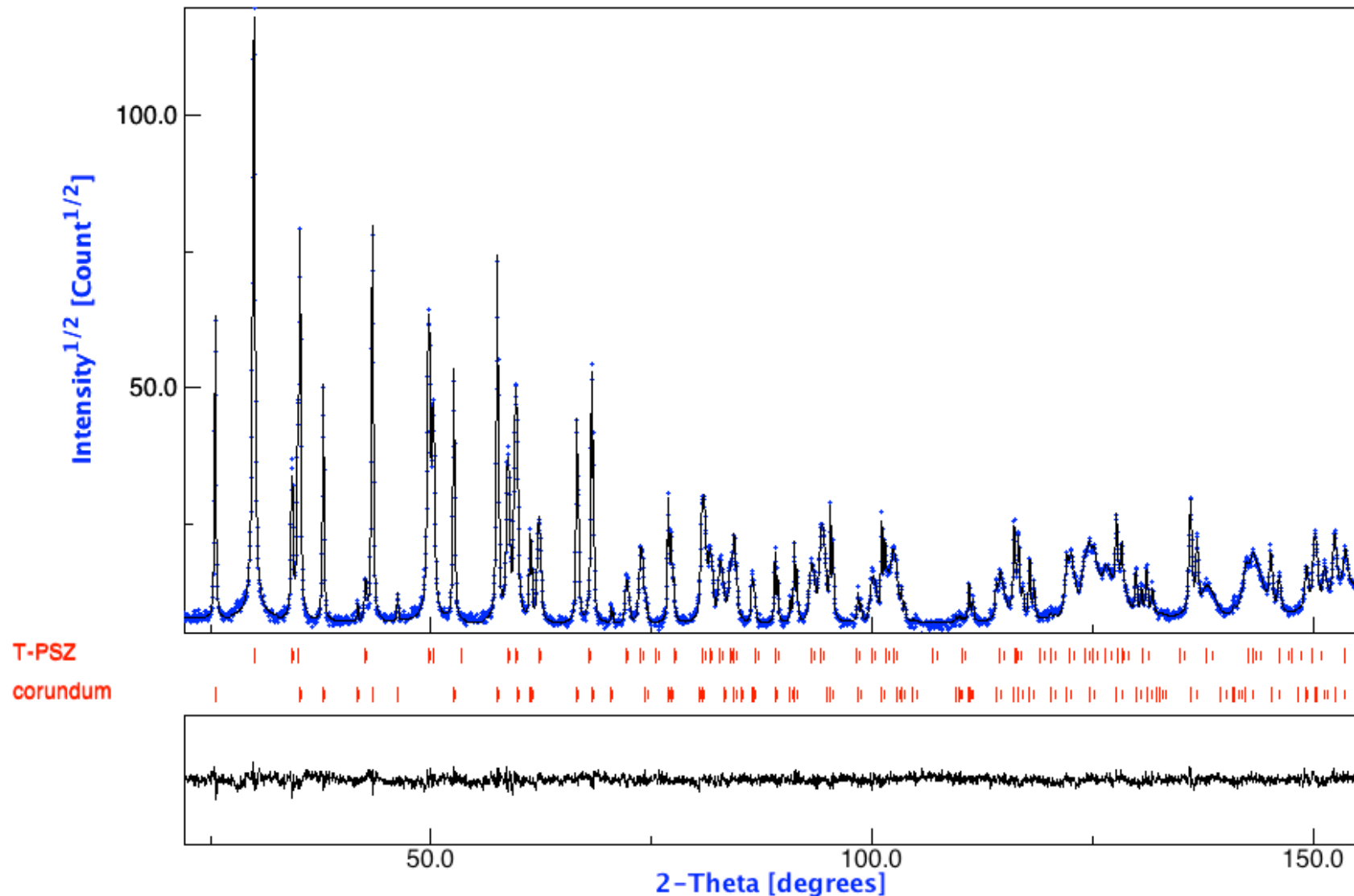
5. Peaks shapes

We add to the refinement also peaks shapes parameters; either the Caglioti parameters (classical programs) or crystallite sizes and microstrains; R_{wp} is now at 9.18 %



6. Crystal structure refinement

Only if the pattern is very good and the phases well defined. We refine separated B factors and only the coordinates that can be refined. Final R_{wp} at 8.86%



Possible cautions

- First get a good experiment/spectrum
- Know your sample as much as possible
- Do not refine too many parameters
- Always try first to manually fit the spectrum as much as possible
- Never stop at the first result
- Look carefully and constantly to the visual fit/plot and residuals during refinement process (no “blind” refinement)
- Zoom in the plot and look at the residuals. Try to understand what is causing a bad fit.
- Do not plot absolute intensities; plot at iso-statistical errors. Small peaks are important like big peaks.
- Use all the indices and check parameter errors.

Finally: What is Physically Meaningful?

Wel (1975)

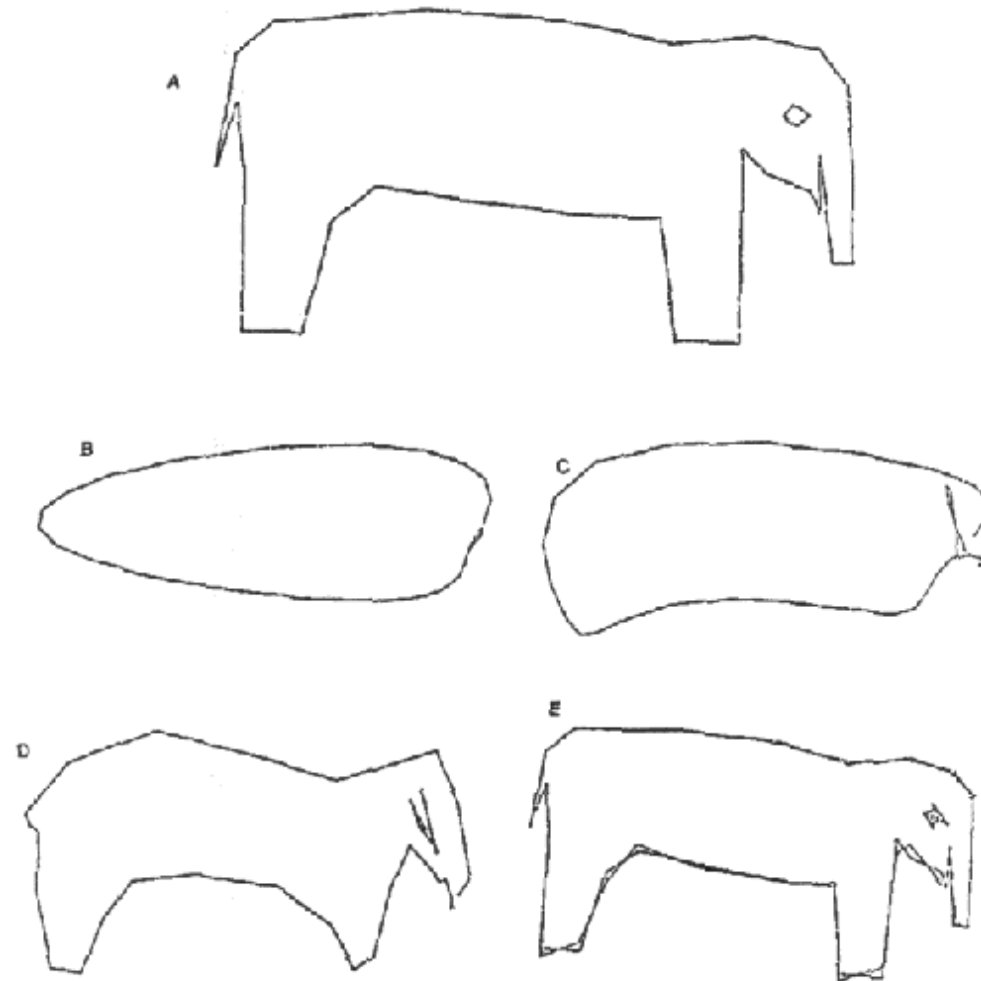


FIGURE 1.2. "How many parameters does it take to fit an elephant?" was answered by Wel (1975). He started with an idealized drawing (A) defined by 36 points and used least squares Fourier sine series fits of the form $x(t) = \alpha_0 + \sum \alpha_i \sin(it\pi/36)$ and $y(t) = \beta_0 + \sum \beta_i \sin(it\pi/36)$ for $i = 1, \dots, N$. He examined fits for $K = 5, 10, 20$, and 30 (shown in B–E) and stopped with the fit of a 30-term model. He concluded that the 30-term model "may not satisfy the third-grade art teacher, but would carry most chemical engineers into preliminary design."